

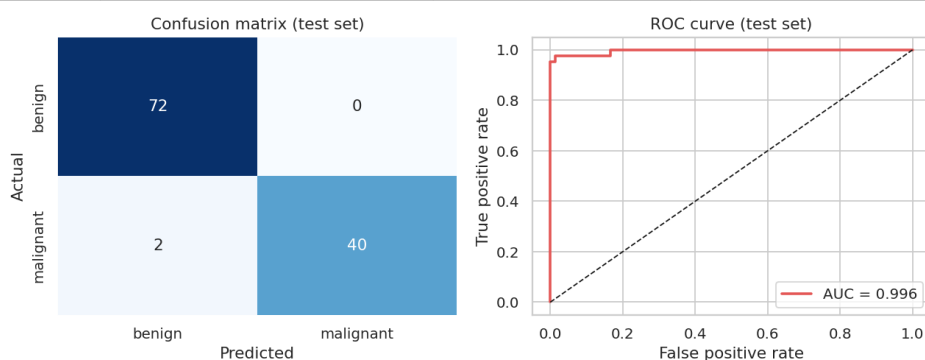
Model Testing Report: Breast-Mass Malignancy Classifier

Final Assignment (Level 0), Part B. Model under test: logistic regression in a full preprocessing pipeline (median imputation, 1st/99th-percentile capping, standardisation, one-hot size group), trained on the public WDBC dataset (569 samples, 30 features). Code and outputs: accompanying Jupyter notebook. Learning project only, not a medical device.

1. Testing strategy

- **Validation plan.** 80/20 stratified hold-out split (455 train / 114 test, malignant share 37.4% / 36.8%), seed 42. Model comparison, preprocessing ablation and tuning used 5-fold stratified cross-validation on the training set only; every preprocessing step sits inside the pipeline so it is refit per fold (no leakage). The test set was used once for final reporting.
- **Metrics and why.** **Recall** on malignant (a false negative is a missed cancer) and **precision** (false alarms) are the key pair; **F1** selects the model; **accuracy** is reported beside a majority-class baseline (about 63%) because it alone is misleading; **ROC-AUC** measures ranking quality independent of threshold; the **confusion matrix** gives raw error counts; **Brier score** checks probability quality. RMSE is a regression metric, so it is not the primary measure here.

Metric (test set, n=114)	Value	Metric	Value
Accuracy	0.982 (95% CI 0.94-1.00)	Recall (malignant)	0.952 (95% CI 0.84-0.99)
Precision (malignant)	1.000	ROC-AUC	0.996
F1 (malignant)	0.976	Errors	0 false positives, 2 false negatives



2. QA functional tests and findings

2.1 Overfitting / underfitting.

Model	Train acc	CV acc	Test acc	Train-CV gap
Majority-class baseline	0.626	0.626	0.632	0.000
Decision stump (depth 1) - underfit probe	0.930	0.901	0.877	0.029
Unpruned decision tree - overfit probe	1.000	0.930	0.921	0.070
FINAL: Logistic Regression (tuned)	0.985	0.982	0.982	0.002

Finding: no overfitting or underfitting in the final model. Train, CV and test accuracy agree (0.985 / 0.982 / 0.982). The probes behave as expected: the stump underfits (0.877 test) and the unpruned tree overfits (gap 0.070). In the learning curve the training and validation curves converge, with validation accuracy still rising slightly, so more data might help a little.

2.2 Fairness / subgroup performance. The dataset has **no demographic attributes**, so a demographic audit is not possible; technical subgroups are used as a proxy.

Subgroup	n	Malignant share	Accuracy (95% CI)	Recall	Precision	FN	FP
large	34	0.88	1.000 (0.90-1.00)	1.000	1.000	0	0
medium	42	0.29	0.952 (0.84-0.99)	0.833	1.000	2	0
small	38	0.00	1.000 (0.91-1.00)	n/a	n/a	0	0
high texture	51	0.65	0.980 (0.90-1.00)	0.970	1.000	1	0
low texture	63	0.14	0.984 (0.92-1.00)	0.889	1.000	1	0

Finding: possible weaker recall on medium-size and low-texture masses, but inconclusive. Recall is 0.833 (medium) vs 1.000 (large), and 0.889 (low texture) vs 0.970 (high). Each gap is driven by one or two patients out of 9 to 12 malignant cases per group; the small-size group has no malignant cases, so its recall is undefined. The data neither confirm nor rule out a disparity; monitor and re-test on a larger sample.

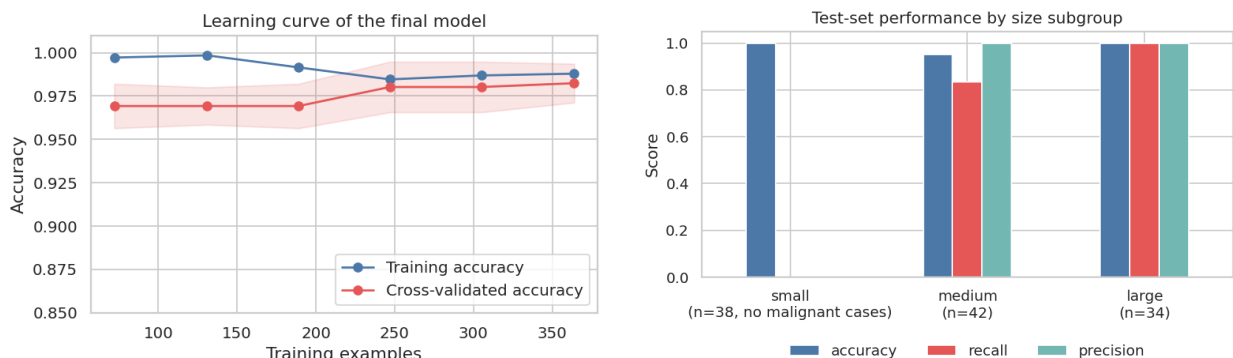


Figure: learning curve of the final model (left) and test-set performance by tumour-size subgroup (right).

2.3 Robustness and threshold.

- **Noise:** accuracy 0.982 falls only to 0.954 with noise equal to half of each feature's std. **Missing values:** imputation prevents crashes; accuracy 0.945 with 20% of values missing.
- **Silent failure on unit errors:** if all inputs are 10x too large the model predicts malignant for everything (accuracy 0.368, recall 1.0) and raises no error. Input range checks are therefore mandatory in deployment.
- **Threshold:** at 0.5 there are 2 missed cancers and 0 false alarms; at 0.4, 1 and 1. The remaining miss has predicted probability about 0.04 and is not recoverable by thresholding. The threshold should be chosen on validation data with clinical input, not on this test set.

3. API / integration tests

The model is not deployed, so a stand-in *POST /predict* function (strict input validation + the saved pipeline) was tested. **12 of 12 cases passed**; output matched direct model output on 25 rows, repeat calls were identical, and latency was about 5 ms per call locally.

Test case	Expected	Result
T1 valid benign sample	200, label=benign	PASS
T2 valid malignant sample	200, label=malignant	PASS
T3 one required field missing	422 error	PASS
T4 empty payload	422 error	PASS
T5 payload is a list, not object	422 error	PASS
T6 number sent as string	422 error	PASS
T7 NaN value	422 error	PASS
T8 negative measurement	422 error	PASS
T9 extra unknown field	200, same label, warning	PASS
T10 every value x100 (implausible)	200 with out-of-range warnings	PASS
T11 boundary: training minimums	200, valid probability	PASS
T12 boundary: training maximums	200, valid probability	PASS

Not yet covered (needed before real deployment): authentication, rate and payload-size limits, concurrency/load tests, model and schema versioning in responses, input/output logging for drift monitoring.

4. Responsible AI appendix

Possible bias source	Mitigation
Sampling and representation: one institution and era; patient demographics and imaging equipment unknown.	Document data provenance; collect multi-site data with metadata; validate externally before any use.
No protected attributes recorded: fairness across demographic groups cannot be audited.	Add and securely handle demographic fields for auditing; set a minimum recall for every subgroup.
Label and measurement bias: diagnoses reflect past clinical practice; image-derived features depend on stain, scanner, software.	Review labels with domain experts; test on data from other labs; monitor feature drift.
Small groups, imbalance, single split: subgroup errors are poorly measured.	Repeated CV, bigger hold-out sets, report confidence intervals; consider re-weighting if a real gap appears.
Automation bias and silent errors: users may over-trust a 98% accurate model; unit errors give confident wrong answers.	Keep a human in the loop; strict input validation and out-of-range warnings; publish a model card with limits and failure modes.
Wrong operating point: default 0.5 threshold ignores unequal cost of errors.	Choose threshold from a cost-based analysis with clinicians, using validation data.

5. Overall conclusion

The model is accurate (98.2% on a small test set), well-behaved on overfitting checks, and robust to noise and missing values. The main risks are not statistical: the **2 missed cancers, unverified fairness** (no demographic data, tiny

subgroups), and **silent failure on badly scaled input**. Recommended next steps: external validation, a cost-based threshold, strict input validation, and subgroup monitoring.